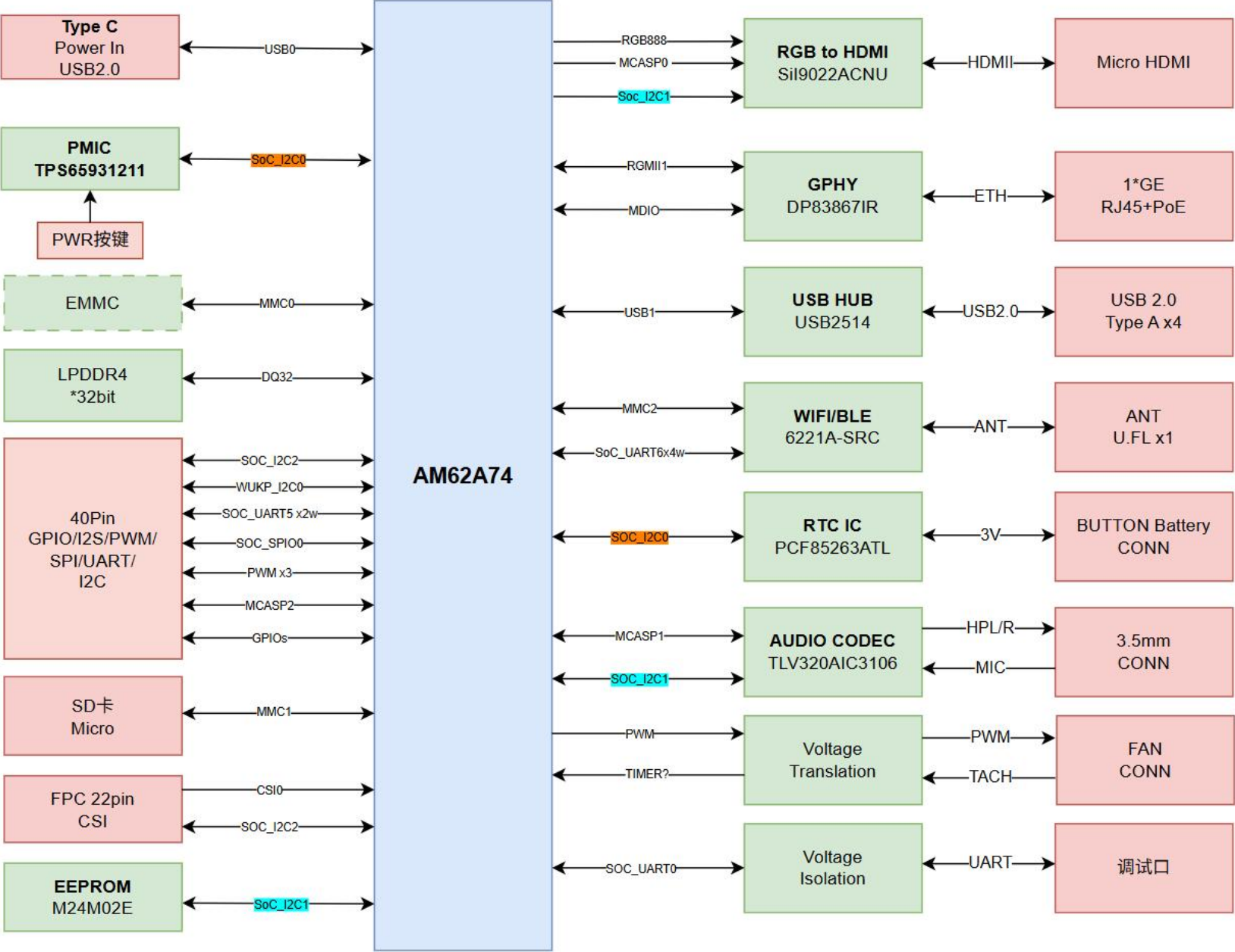
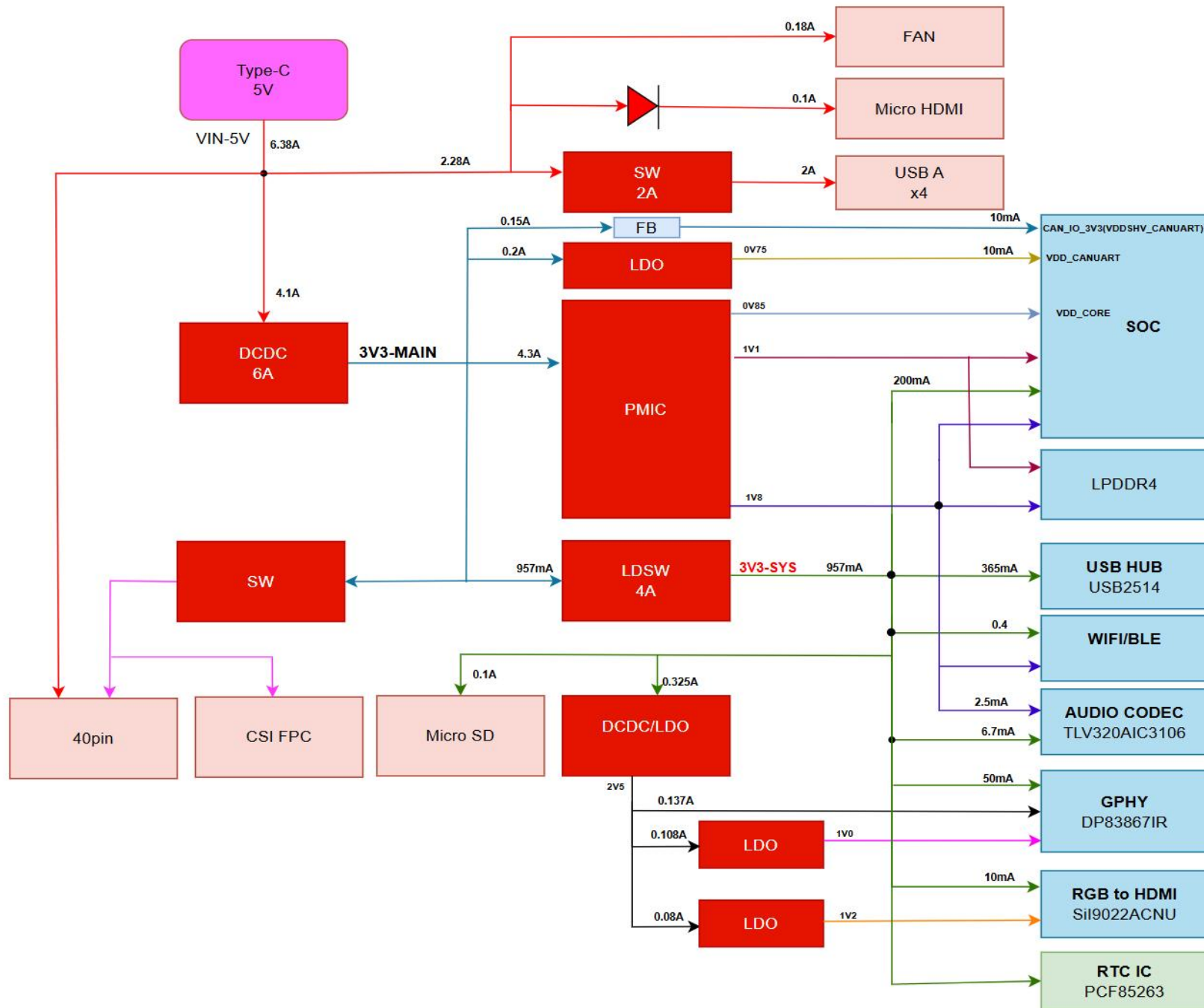


BLOCK DIAGRAM AM62A



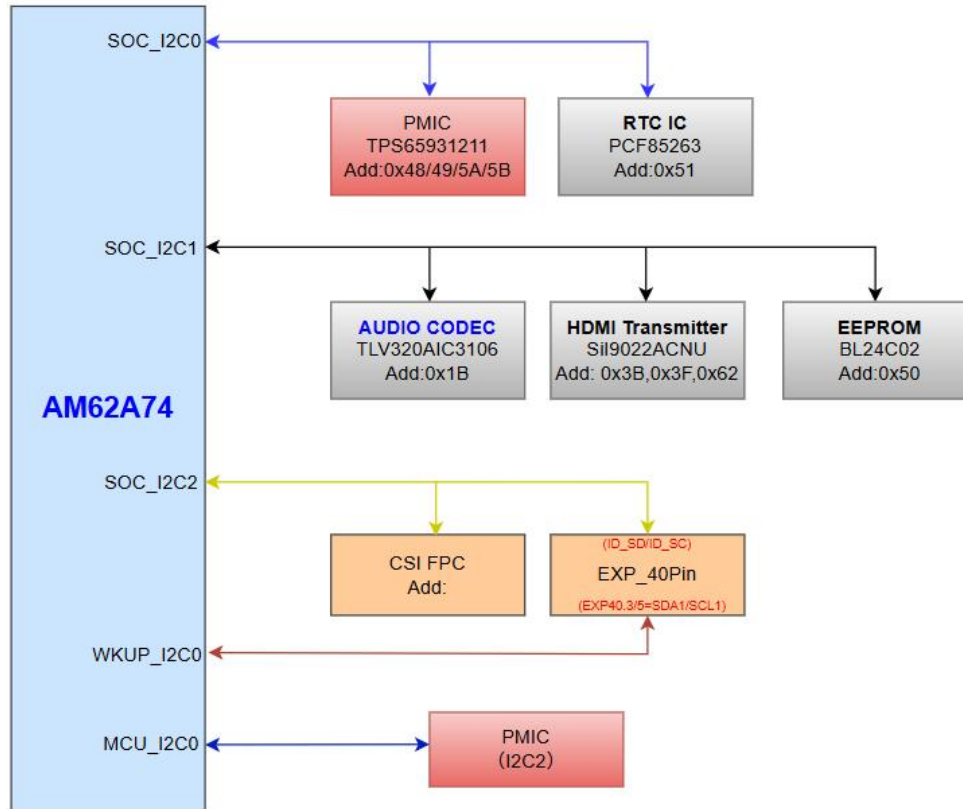
POWER BLOCK DGM



NOR

 Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	02. POWER BLOCK DGM
Date:	Wednesday, April 01, 2026
Rev:	V1.0
Designed by:	
Reviewed by:	
Sheet:	2 of 22

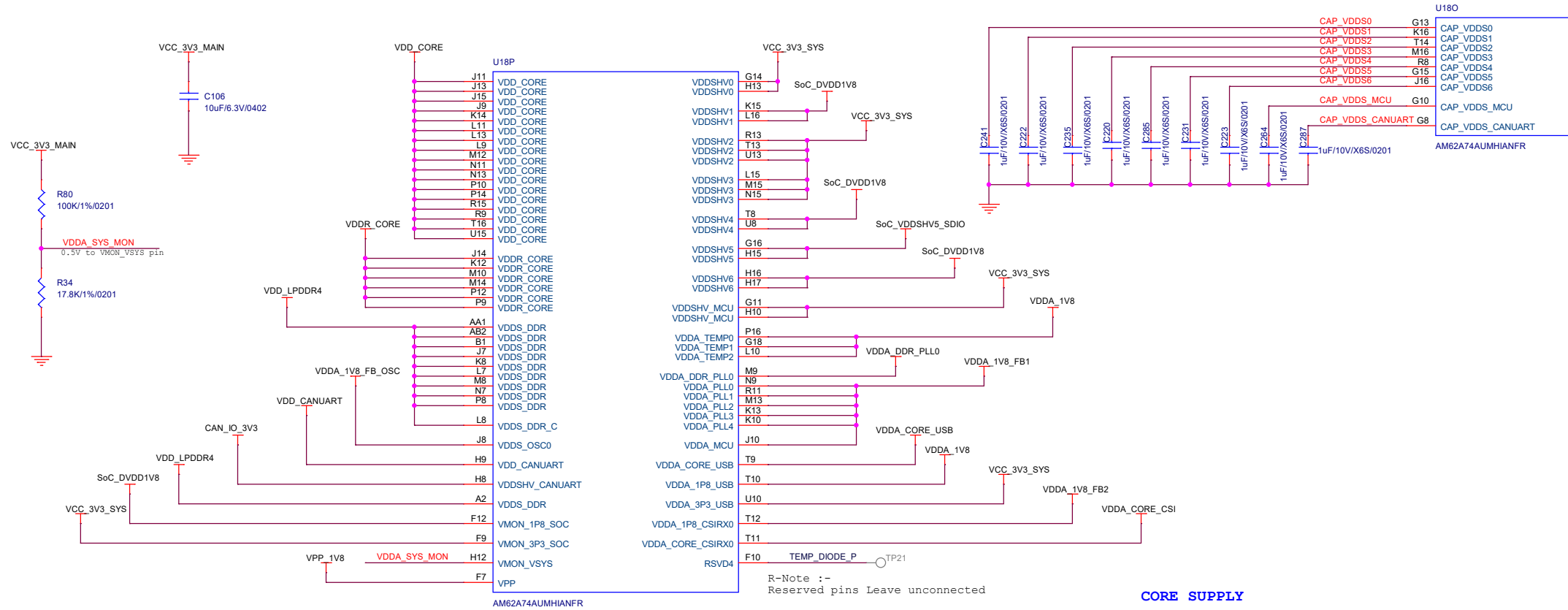
I2C TREE



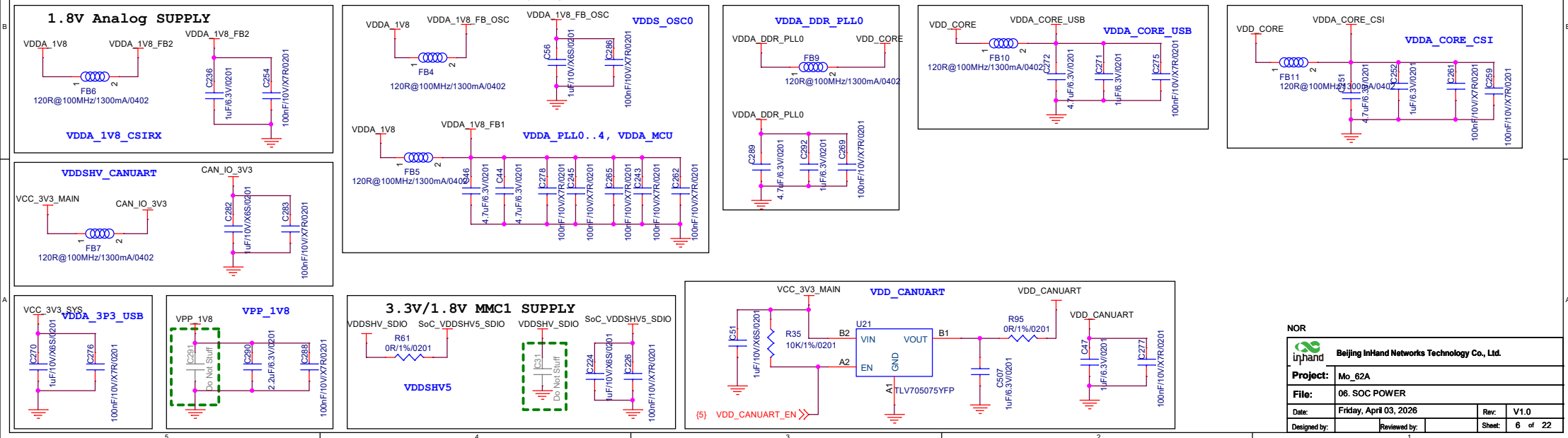
NOR

		Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A		
File:	03. I2C TREE		
Date:	Wednesday, April 01, 2026		Rev: V1.0
Designed by:		Reviewed by:	Sheet: 3 of 22

SOC POWER

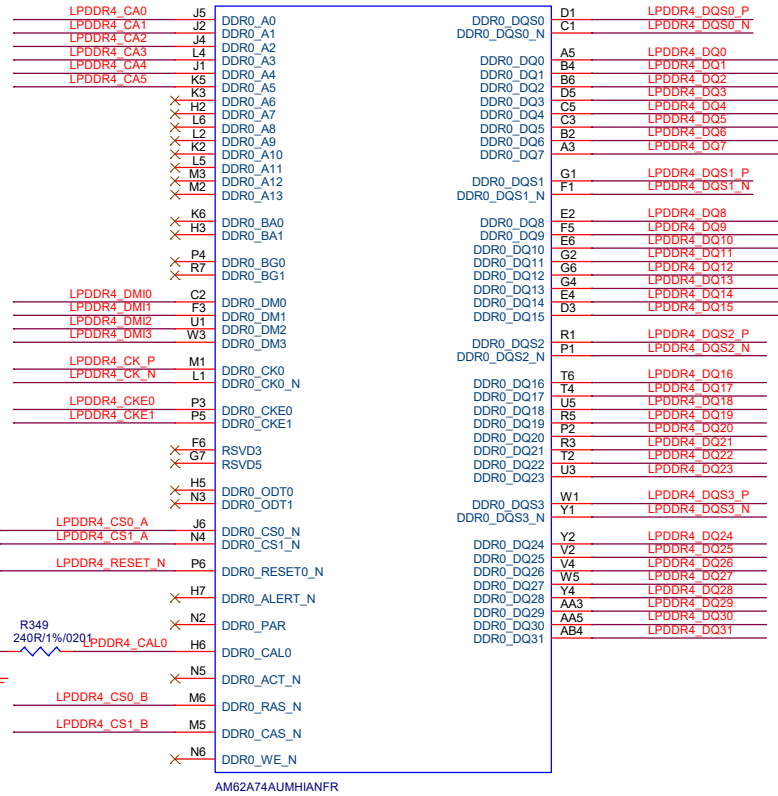


CORE SUPPLY

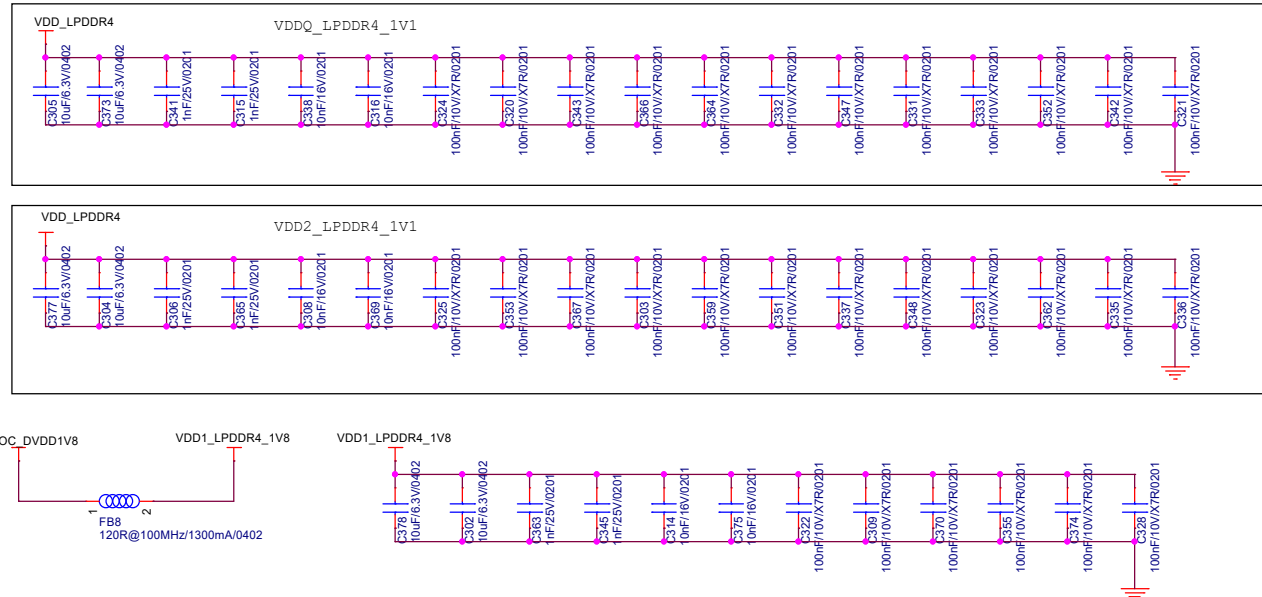


SOC LPDDR4 INTERFACE

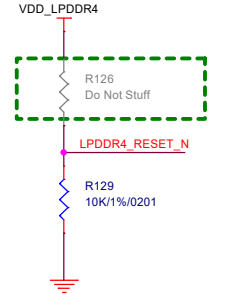
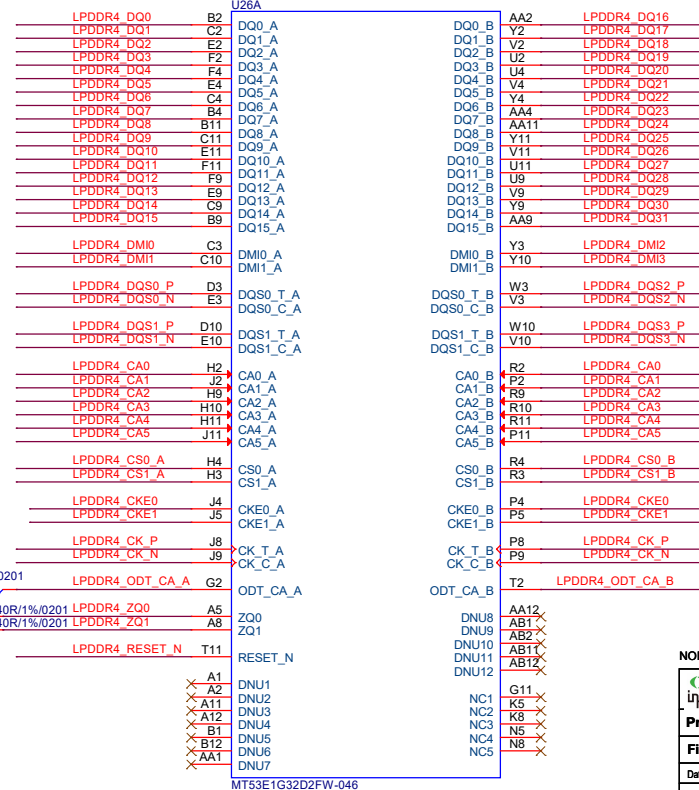
U18G



LPDDR4 POWER DECAPS



LPDDR4 DEVICE



R-Note :-
Pull-down is populated

NOR

inhand

Beijing InHand Networks Technology Co., Ltd.

Project: Mo_62A

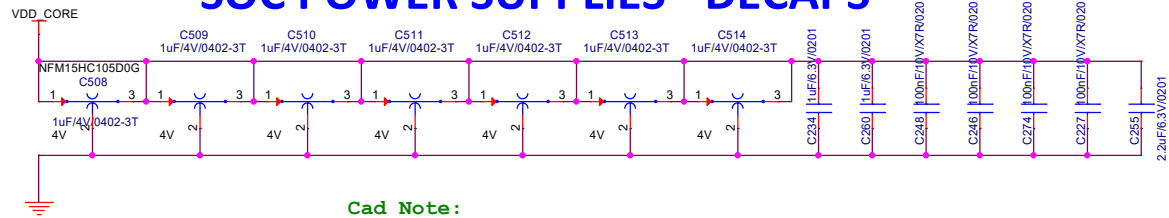
File: 08. LPDDR4 INTERFACE

Date: Friday, April 03, 2026

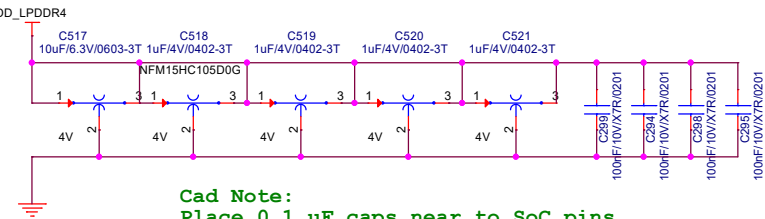
Designed by: Reviewed by: Rev: V1.0

Sheet: 8 of 22

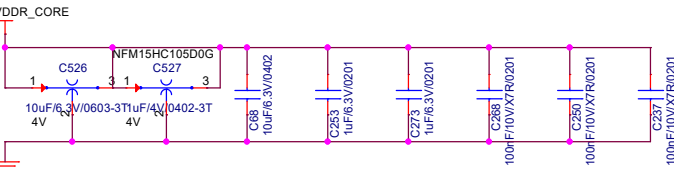
SOC POWER SUPPLIES - DECAPS



Cad Note:
Place 0.1 uF caps near to SoC pins

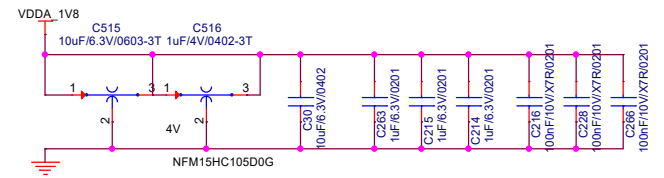
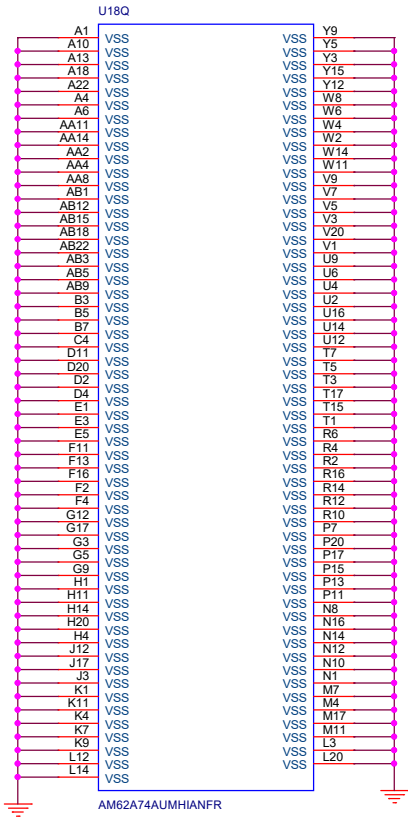


Cad Note:
Place 0.1 uF caps near to SoC pins

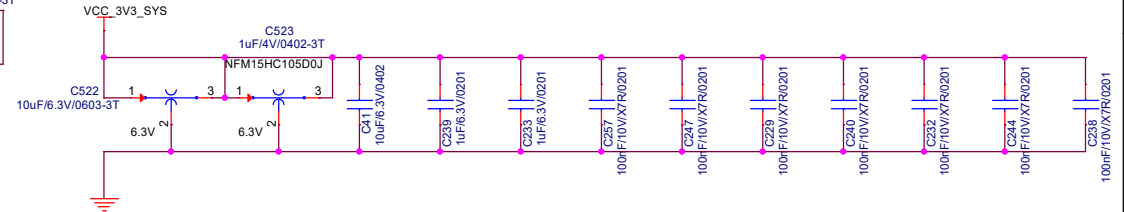


Cad Note:
Place 0.1 uF caps near to SoC pins

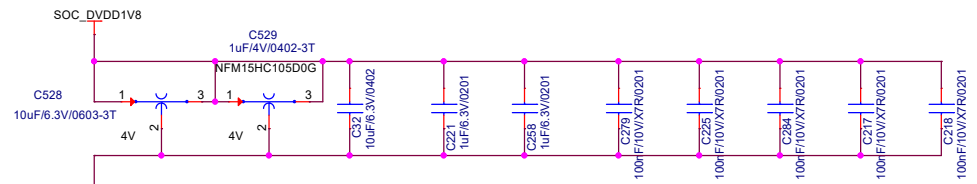
SOC VSS



Cad Note:
Place 0.1 uF caps near to SoC pins



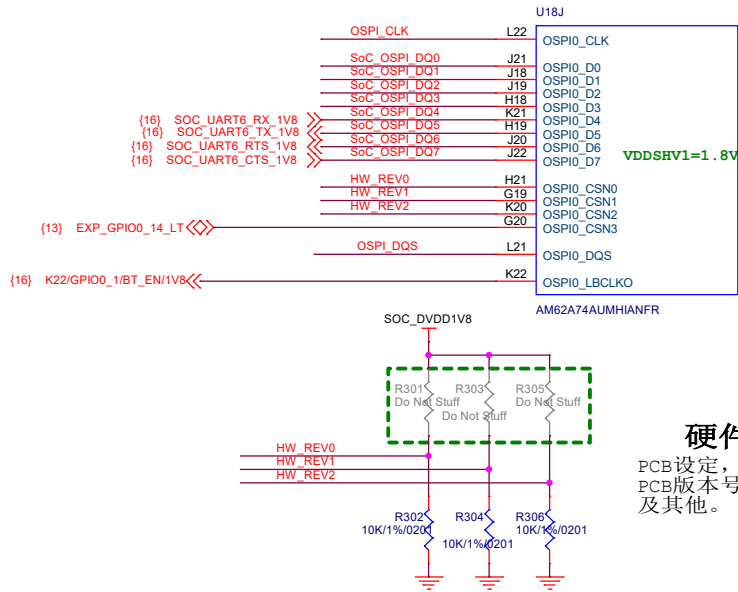
Cad Note:
Place 0.1 uF caps near to SoC pins



Cad Note:
Place 0.1 uF caps near to SoC pins

NOR			
 Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	07. SOC POWER CAPS		
Date:	Friday, April 03, 2026	Rev:	V1.0
Designed by:		Reviewed by:	
		Sheet:	7 of 22

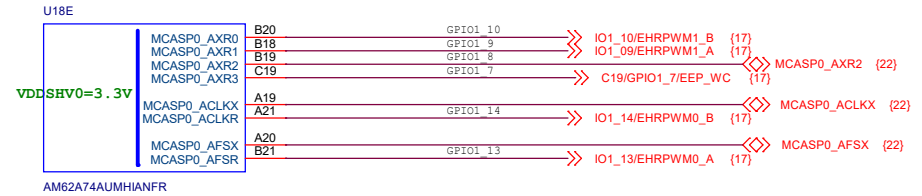
SOC OSPI INTERFACE



硬件设置

PCB设定，用于设置
PCB版本号、DDR型号、
及其他。

SOC MCASP0 INTERFACE



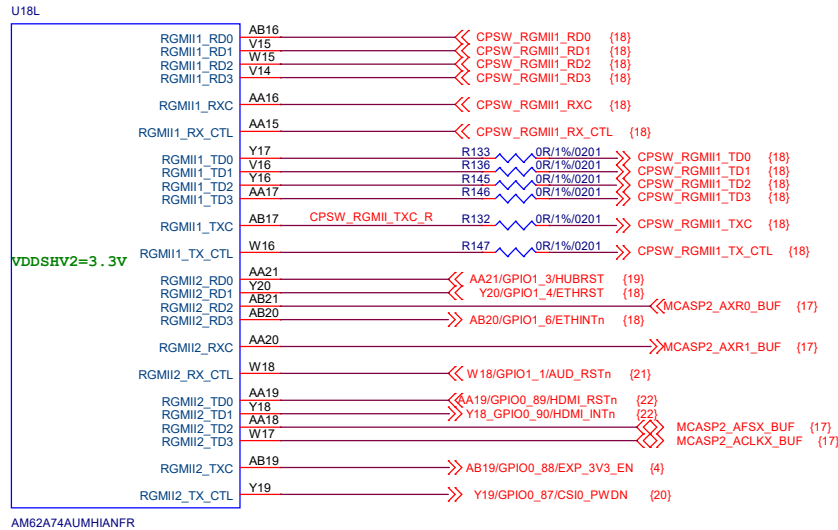
GPIO1_13-----SOC_SPI2_CS0

GPIO1_14-----SOC_SPI2_CLK

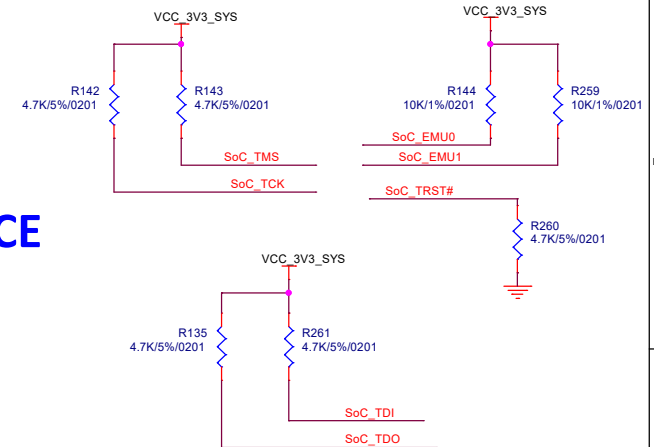
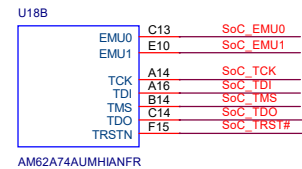
GPIO1_7 -----SOC_SPI2_D0

GPIO1_8 -----SOC_SPI2_D1

SOC MAC INTERFACE



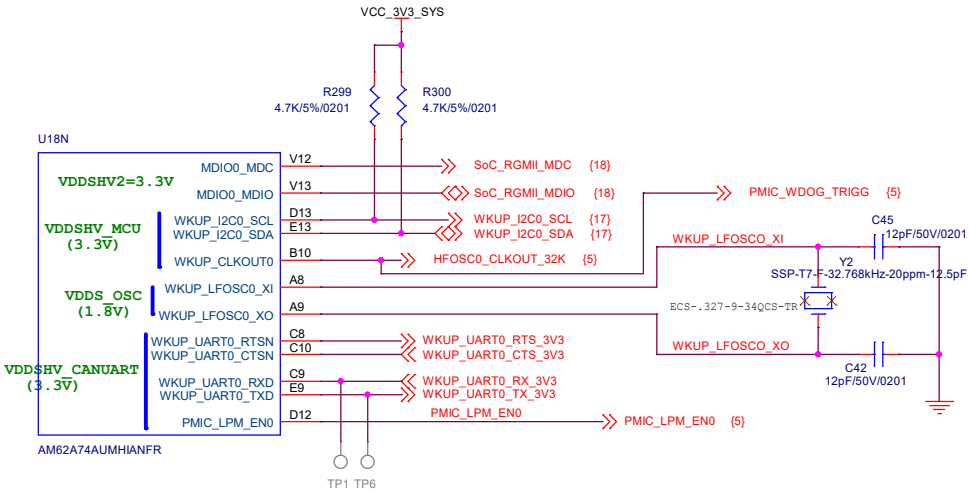
SOC JTAG INTERFACE



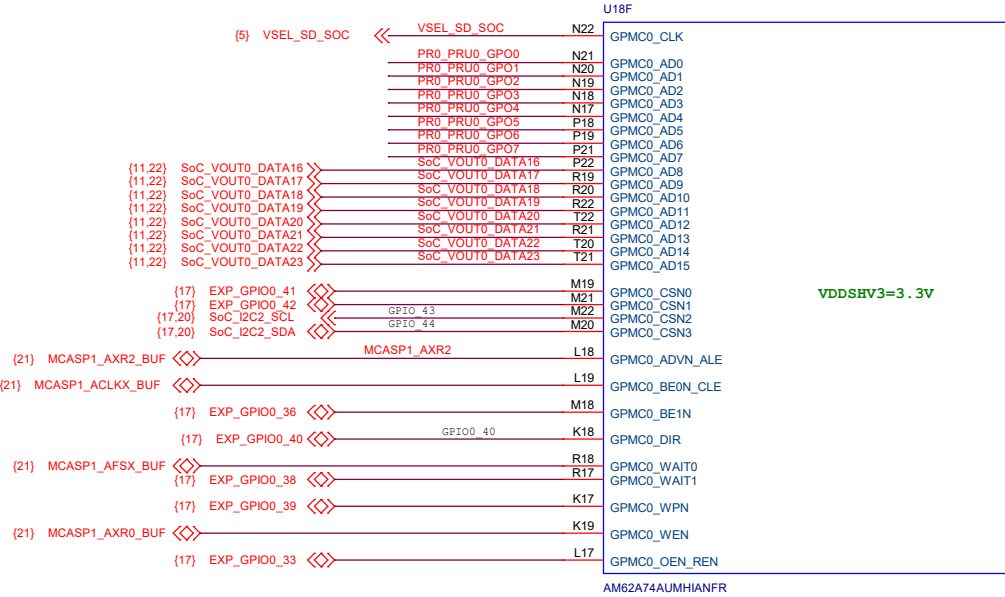
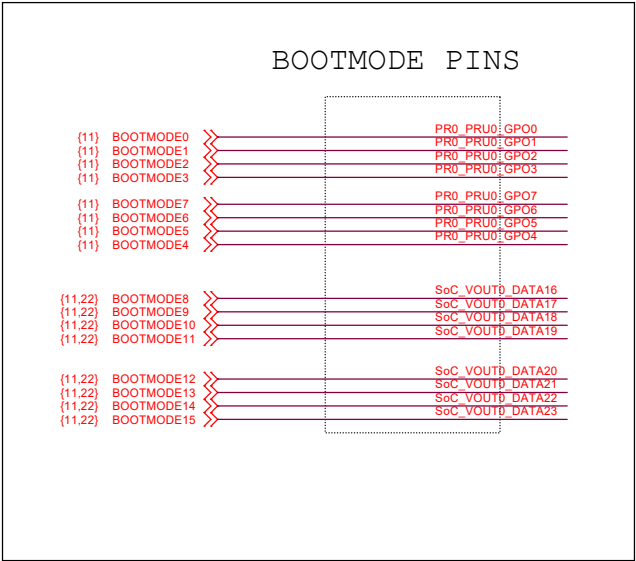
NOR

 Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	09_SOC_MAC,MCASP0,OSPI,JTAG
Date:	Friday, April 03, 2026
Designed by:	Reviewed by:
Rev:	V1.0
Sheet:	9 of 22

SOC WKUP DOMAIN



SOC GPMC INTERFACE



BOOTMODE CONFIGURATION RESISTORS

BOOTMODE7 (10)
BOOTMODE6 (10)
BOOTMODE5 (10)
BOOTMODE4 (10)
BOOTMODE3 (10)
BOOTMODE2 (10)
BOOTMODE1 (10)
BOOTMODE0 (10)

BOOTMODE15 (10,22)
BOOTMODE14 (10,22)
BOOTMODE13 (10,22)
BOOTMODE12 (10,22)
BOOTMODE11 (10,22)
BOOTMODE10 (10,22)
BOOTMODE9 (10,22)
BOOTMODE8 (10,22)

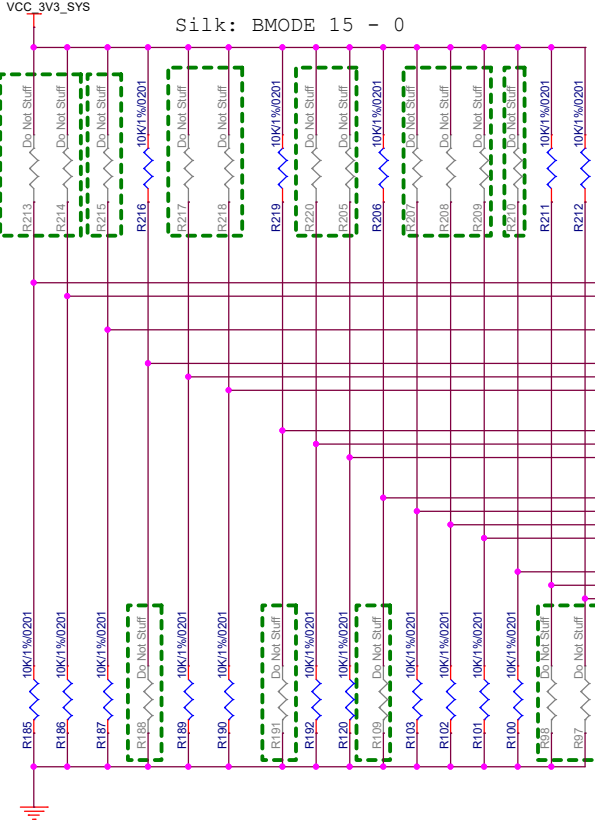
D-Note :-
VCC3V3_XDS TA supply is used to support test automation.
Connect SOC_DVDD3V3 in the custom board design when buffers are not used

D-Note :-
it is not recommended to leave any of the the bootmode pin open including reserved pins

D-Note :-
Dip switch is optional and used on the SK for ease of configuration
A pullup or pulldown resistor can be used to set the BOOTMODE configuration
Provide provision for Pullup and Pulldown resistors for the bootmode pins that have configuration capability

D-Note :-
When dip switches are used on custom board, an external ESD protection may be required if the DIP switches are expected to be configured in an uncontrolled ESD environment

D-Note :-
When DIP switch are used, reduce the resistor values used for the divider to 47K and 470R maintaining the ratio



Silk: BMODE 15 - 0

BOOTMODE15	0	MCU_BOOTMODE[15:14]: Reserved pins.
BOOTMODE14	0	
BOOTMODE13	0	MCU_BOOTMODE[13]: Backup Boot Mode Config, set to '0':RGMI with internal TX delay.
BOOTMODE12	1	MCU_BOOTMODE[12:10]: Backup Boot Mode, set to '100' for Backup boot is Ethernet.
BOOTMODE11	0	
BOOTMODE10	0	
BOOTMODE9	1	MCU_BOOTMODE[9:7]: Primary Boot Mode Config, set to '100'. B9=1: MMC Port 1 (4 bit width).
BOOTMODE8	0	B8=x: Reserved.
BOOTMODE7	0	B7=0: Filesystem mode.
BOOTMODE6	1	
BOOTMODE5	0	
BOOTMODE4	0	MCU_BOOTMODE[6:3]: Primary Boot Mode, set to '1000' for primary boot is MMCSD Boot (SD Card).
BOOTMODE3	0	
BOOTMODE2	0	
BOOTMODE1	1	BOOTMODE[2:0]: PLL Config, set to '011' for 25MHz input frequency.
BOOTMODE0	1	

BOOT MODES SUPPORTED:
Primary Boot: SD CARD
Backup Boot: Ethernet.

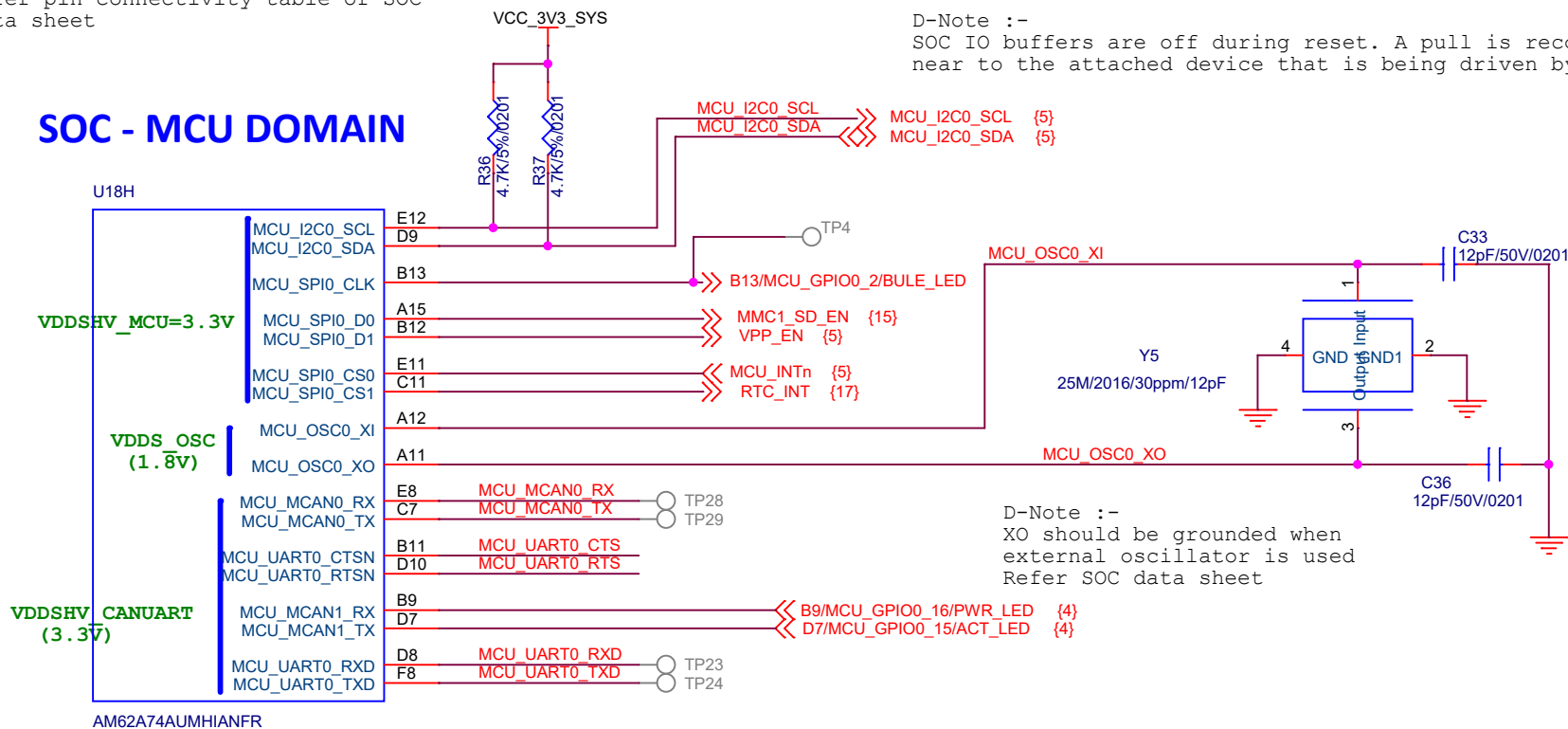
BOOT MODES SUPPORTED

1. OSPI
2. MMC1 - SD CARD
3. UART
4. eMMC
5. ETHERNET
6. USB0 DFU
7. USB0 MS

D-Note :-
A pullup is recommended for Open-drain output type I2C interfaces irrespective of the IO configuration Refer pin connectivity table of SOC data sheet

D-Note :-
Open-drain output type buffer I2C interfaces have slew rate requirement when pulled to 3.3V An RC is recommended for slew rate control. Refer SK-AM62P-LP schematics

D-Note :-
SOC IO buffers are off during reset. A pull is recommended near to the attached device that is being driven by the SOC IO



D-Note :-
XO should be grounded when external oscillator is used Refer SOC data sheet

D-Note :-
Connect the 25 MHz crystal directly to the SOC Xi and Xo pins (No Series or parallel resistors are recommended). The internal oscillator implements AGC (Automatic Gain Control) for amplitude control. Match the SOC and the EPHY crystal specs

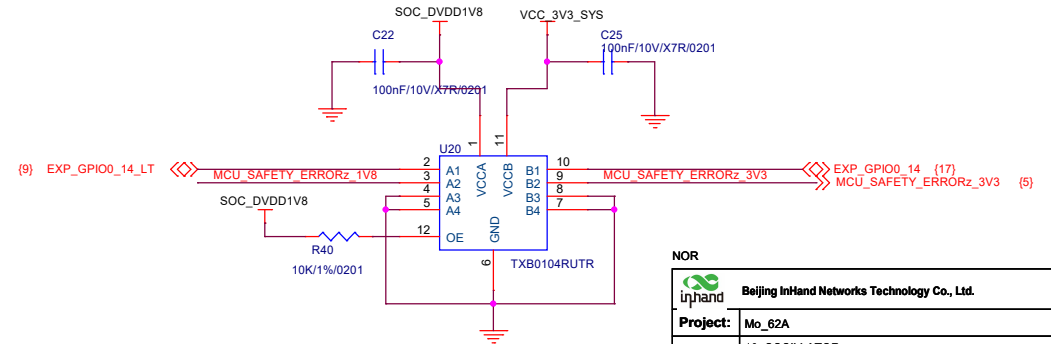
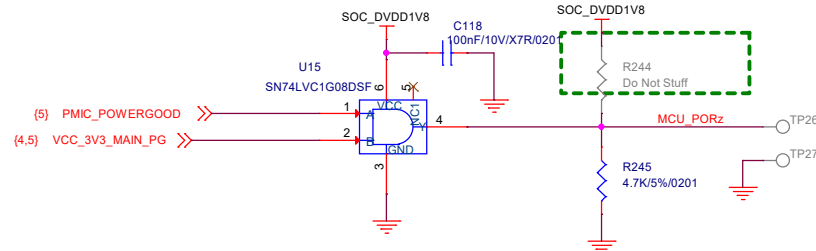
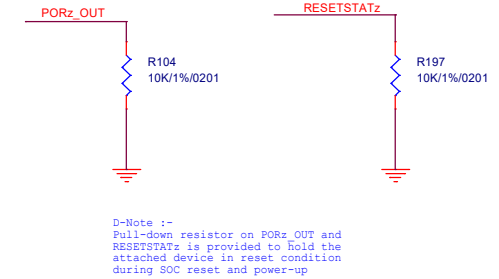
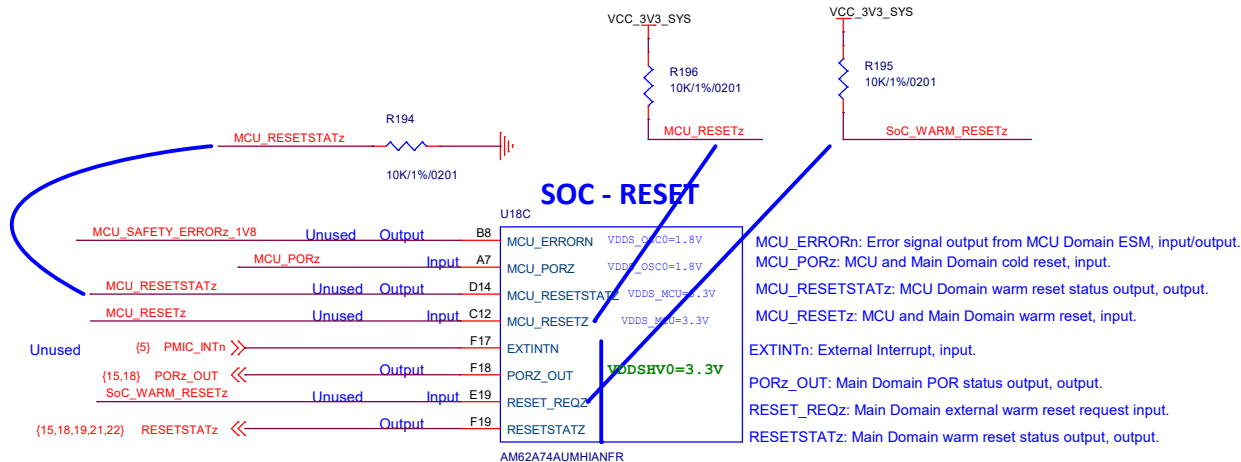
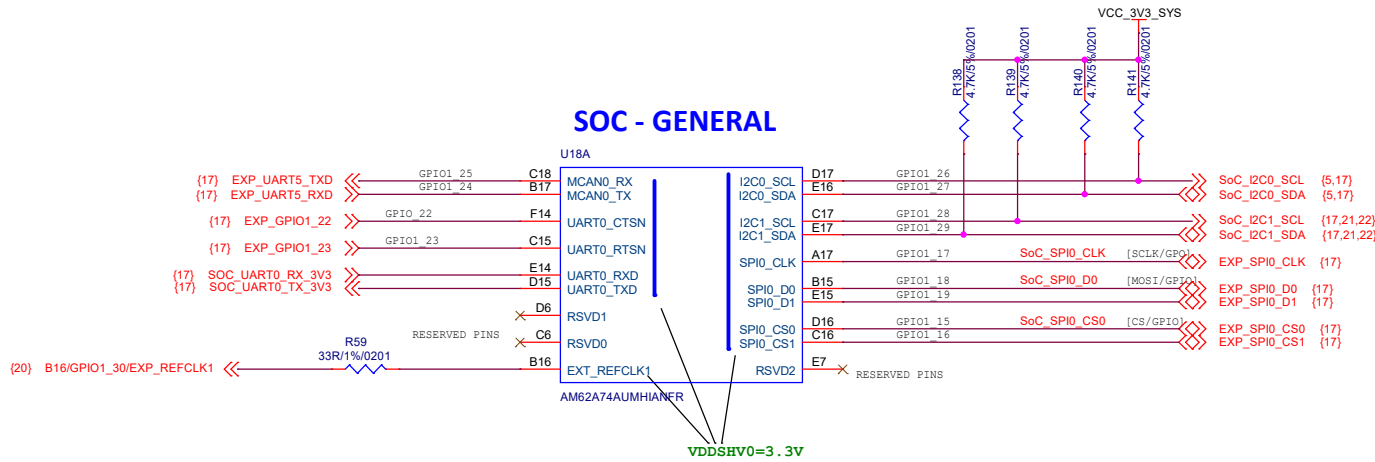
D-Note :-
Refer Applications, Implementation, and Layout section of the data sheet for clock routing guidelines as below:
Clock Routing Guidelines
Oscillator Routing

D-Note :-
No HFOSC0 registers are required to be changed. These registers should remain in their default state. Select the appropriate crystal circuit components that are compliant to the values defined in the MCU_OSC0 Crystal Circuit Requirements table. Read the Load Capacitance and Shunt Capacitance sections to select the appropriate crystal circuit components.

D-Note :-
MCU_OSC0 has been validated only with a 25 MHz clock source, so that is the only frequency supported. The datasheet shows MCU_OSC0 not starting until after the core voltage because there are some cases where the oscillator may not start until VDD_CORE is valid. In most cases it will start as early as VDDS_OSC0, but this may not always be the case. This diagram in the datasheet is showing the maximum start-up time, which must include the case where the delay is based on VDD_CORE being valid.

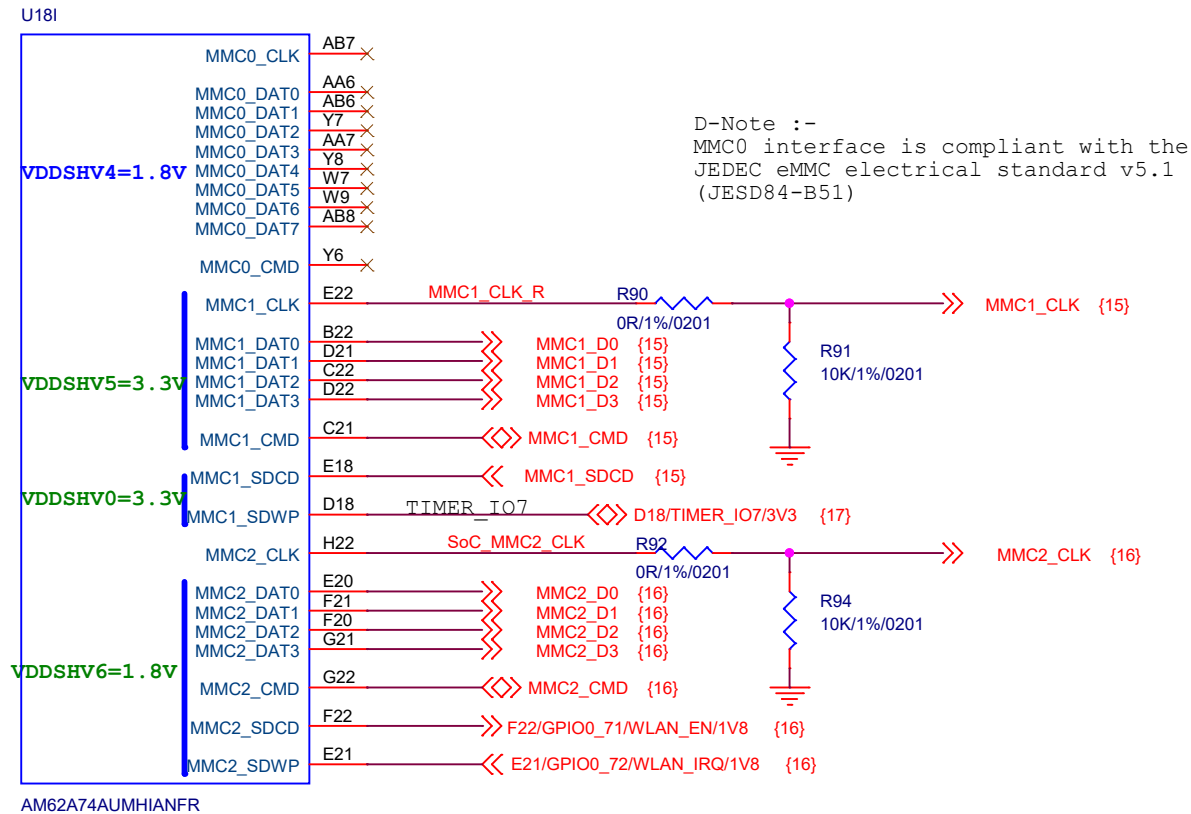
NOR

 Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	12. MCU HEADER
Date:	Wednesday, April 01, 2026
Designed by:	Reviewed by:
Rev:	V1.0
Sheet:	12 of 22



Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	13. OSCILLATOR		
Date:	Friday, April 10, 2026	Rev:	V1.0
Designed by:		Reviewed by:	
Sheet:	13 of 22		

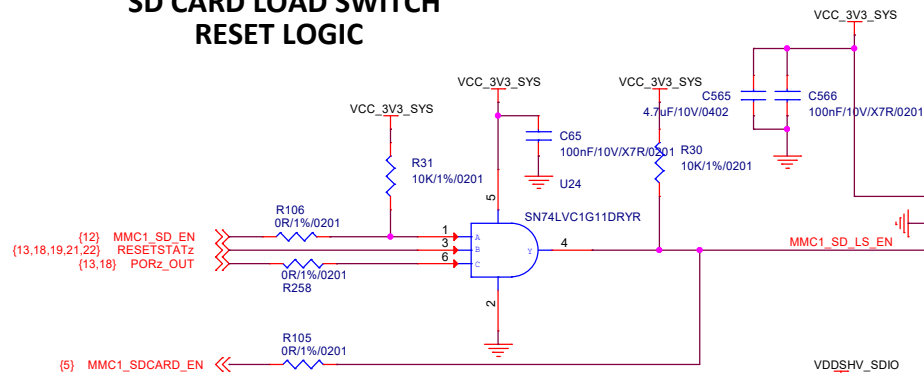
SOC - MMC Interface



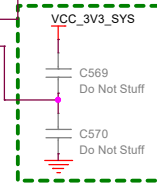
NOR

 Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	14. MMCx		
Date:	Friday, April 03, 2026		Rev: V1.0
Designed by:		Reviewed by:	
		Sheet:	14 of 22

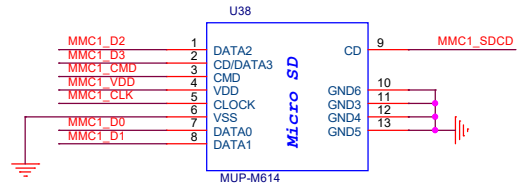
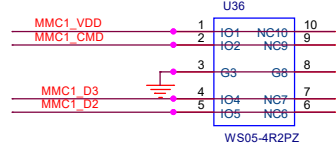
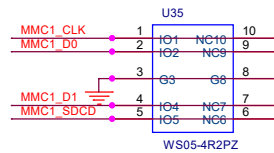
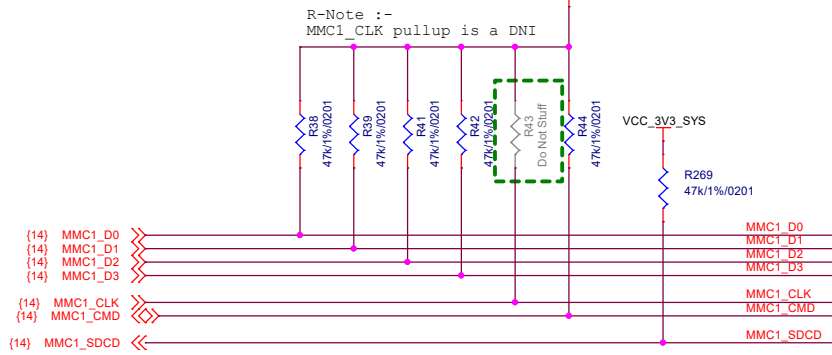
SD CARD LOAD SWITCH RESET LOGIC



LOAD SWITCH



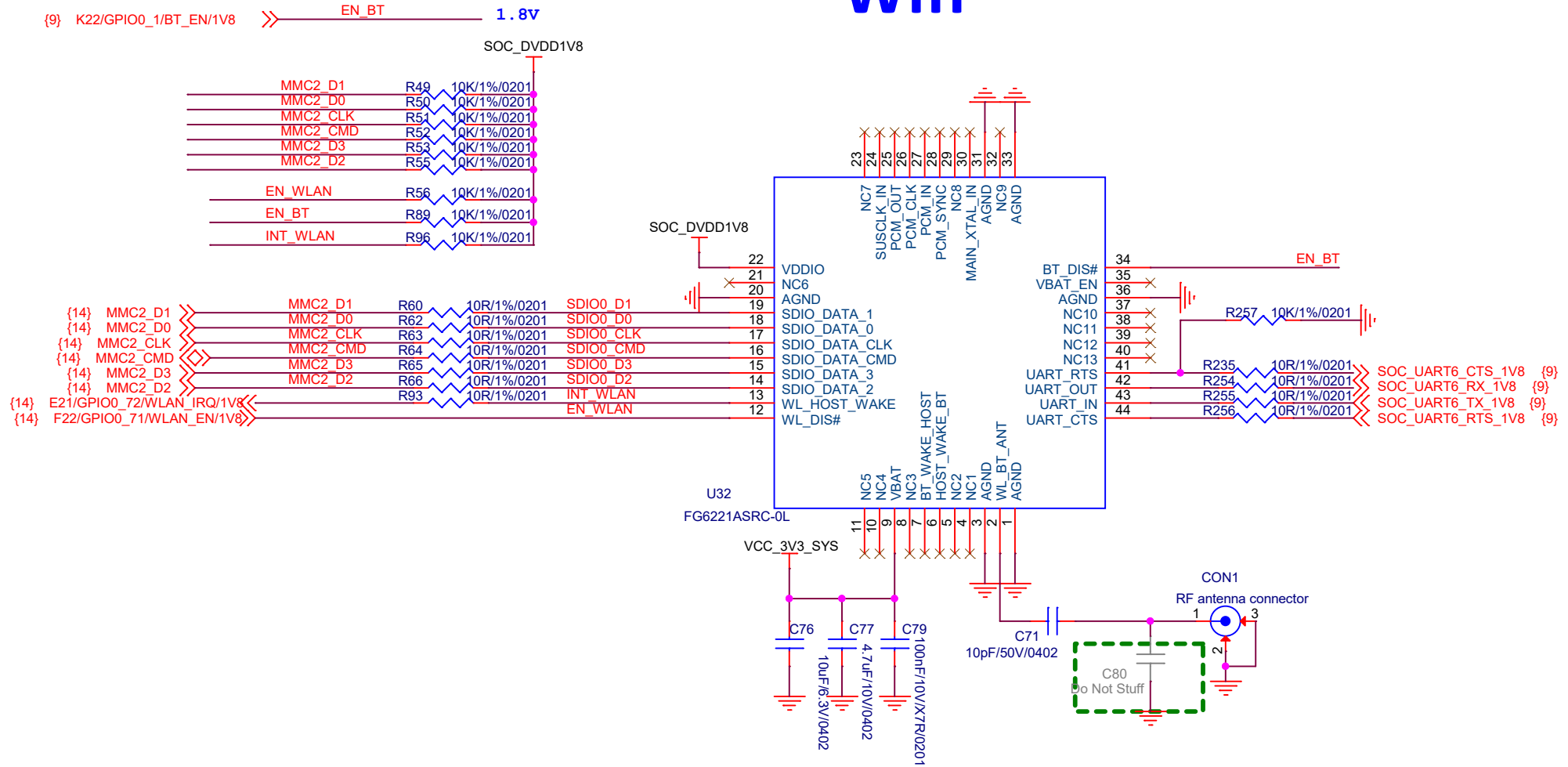
D-Note :-
For UHS-I operation, the pullups are recommended to be connected to the 3.3V/1.8V switched LDO output



NOR

inhand Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	15. SD CARD INTERFACE
Date:	Thursday, April 09, 2026
Designed by:	Reviewed by:
Rev:	V1.0
Sheet:	15 of 22

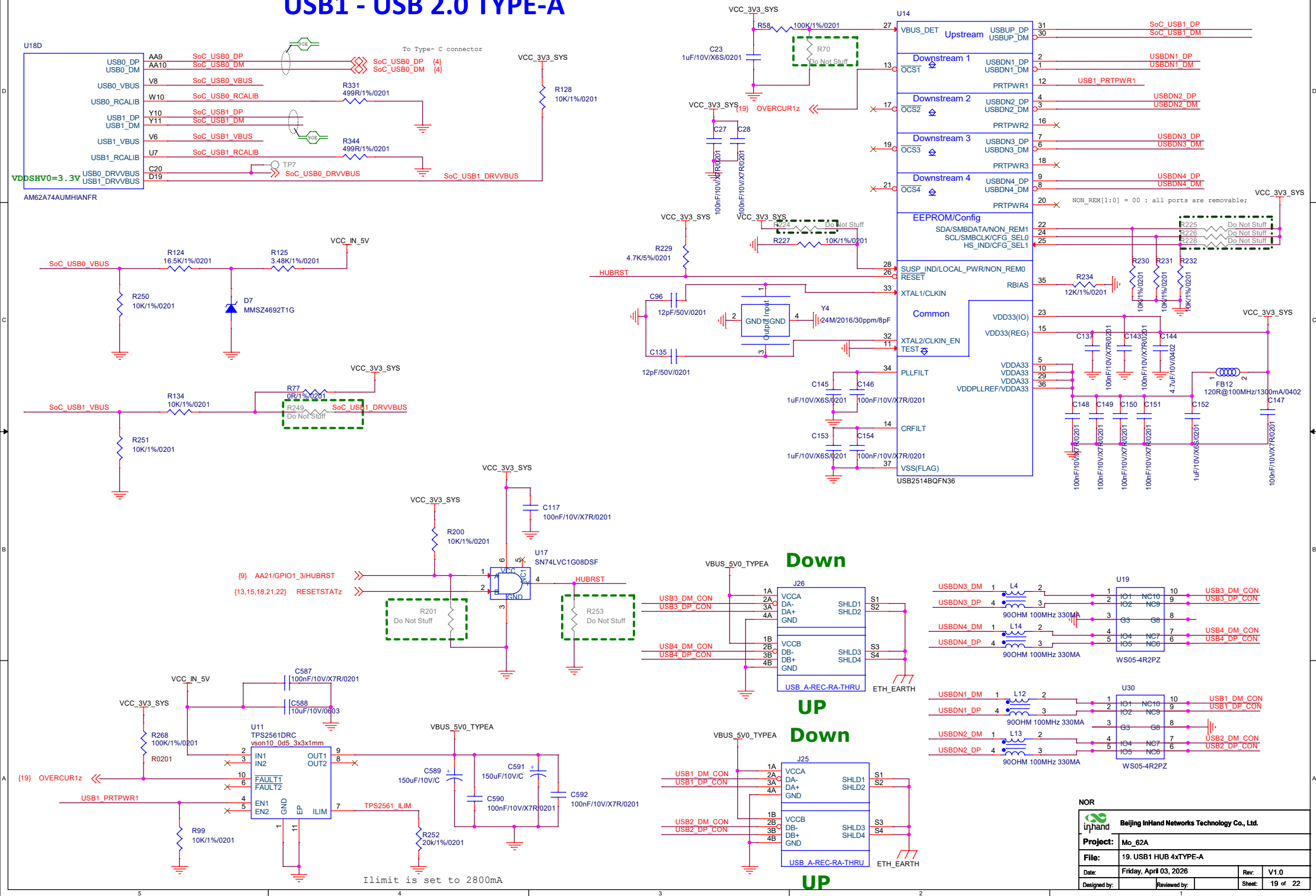
Wifi



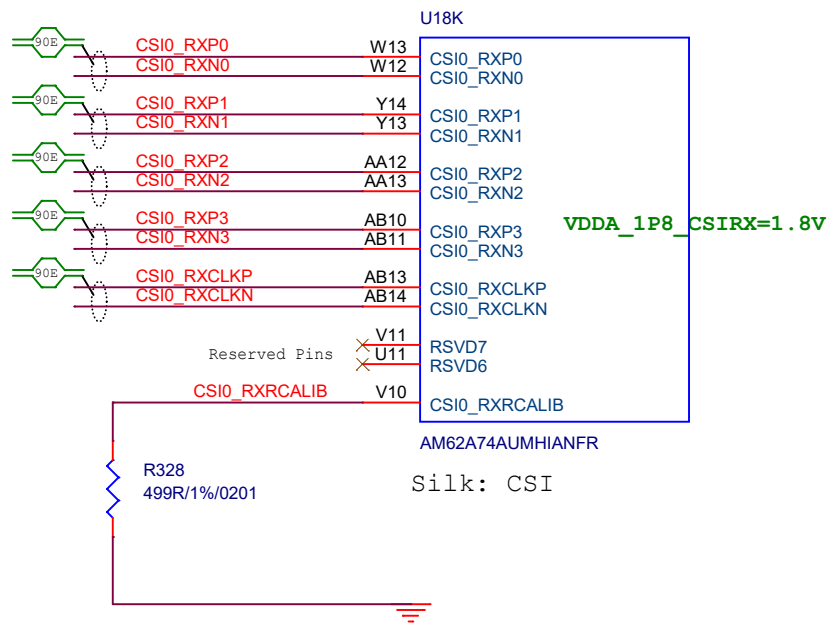
NOR

 Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	16. WIFI/BLE		
Date:	Friday, April 03, 2026		Rev: V1.0
Designed by:		Reviewed by:	Sheet: 16 of 22

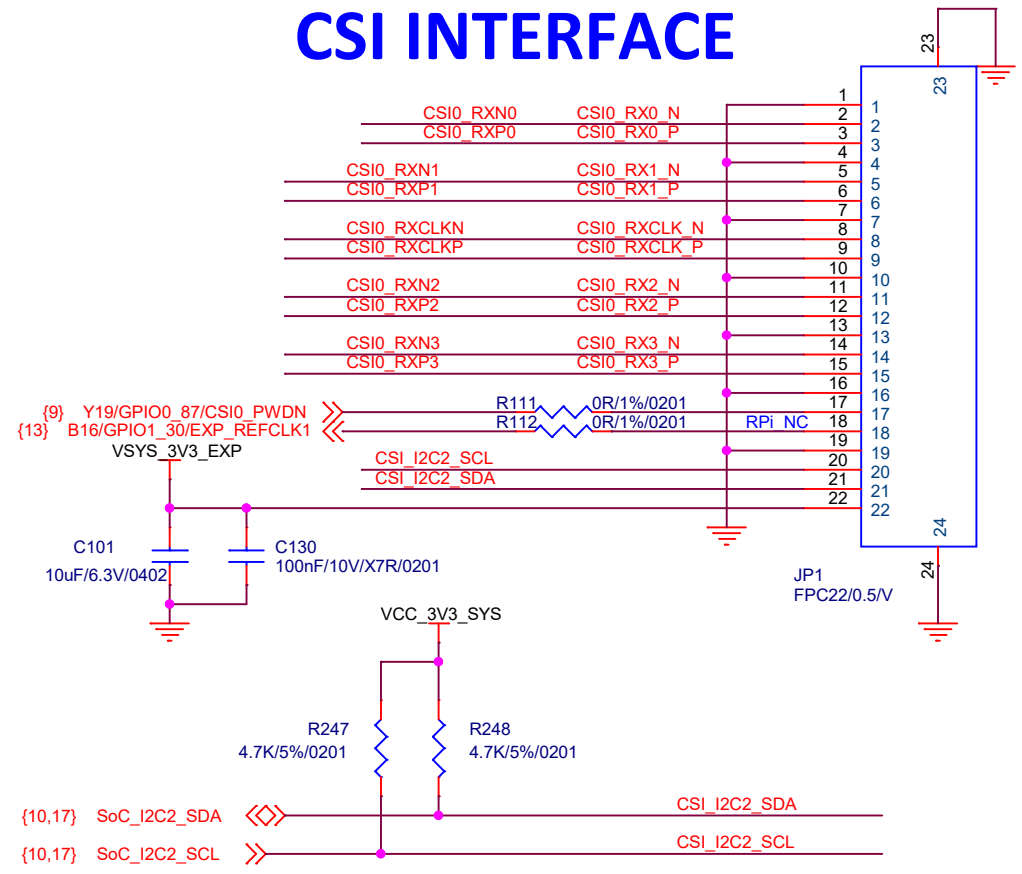
USB1 - USB 2.0 TYPE-A



SOC CSI INTERFACE



CSI INTERFACE

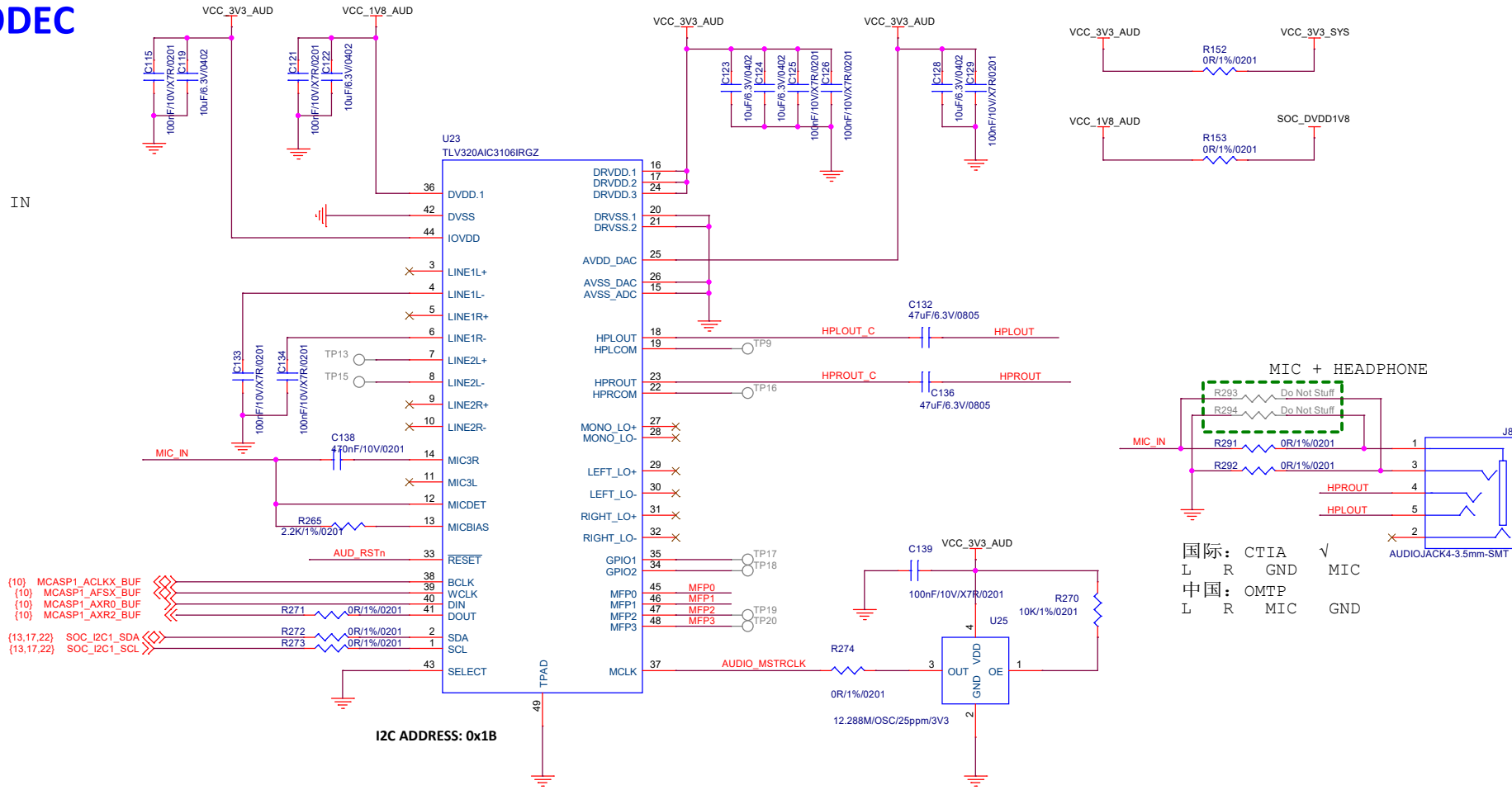


Raspberry Pi Camera Connector x4 Lane

inhand Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	20. CSI INTERFACE		
Date:	Friday, April 03, 2026	Rev:	V1.0
Designed by:		Reviewed by:	
		Sheet:	20 of 22

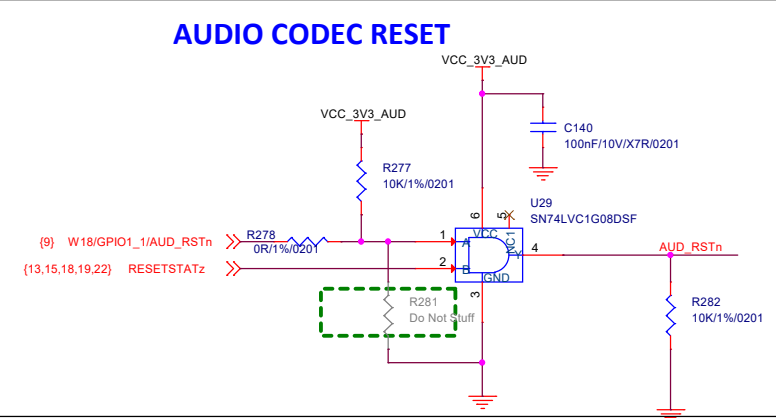
AUDIO CODEC

Silk: LINE IN

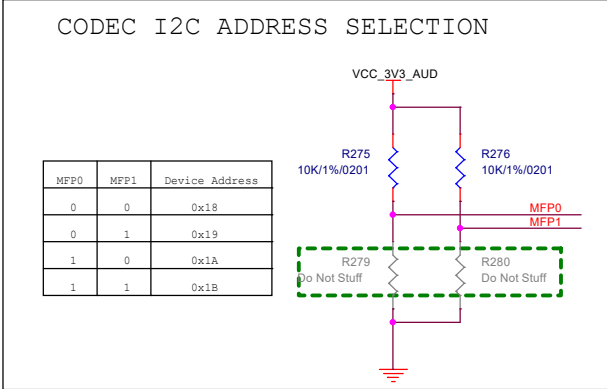


国际: CTIA
L R GND MIC
中国: OMTP
L R MIC GND

AUDIO CODEC RESET



CODEC I2C ADDRESS SELECTION



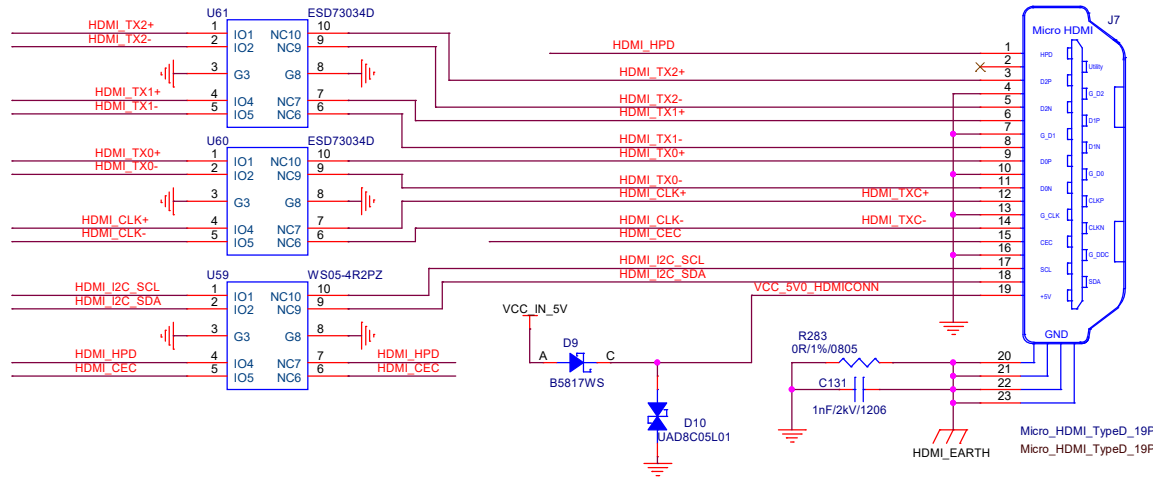
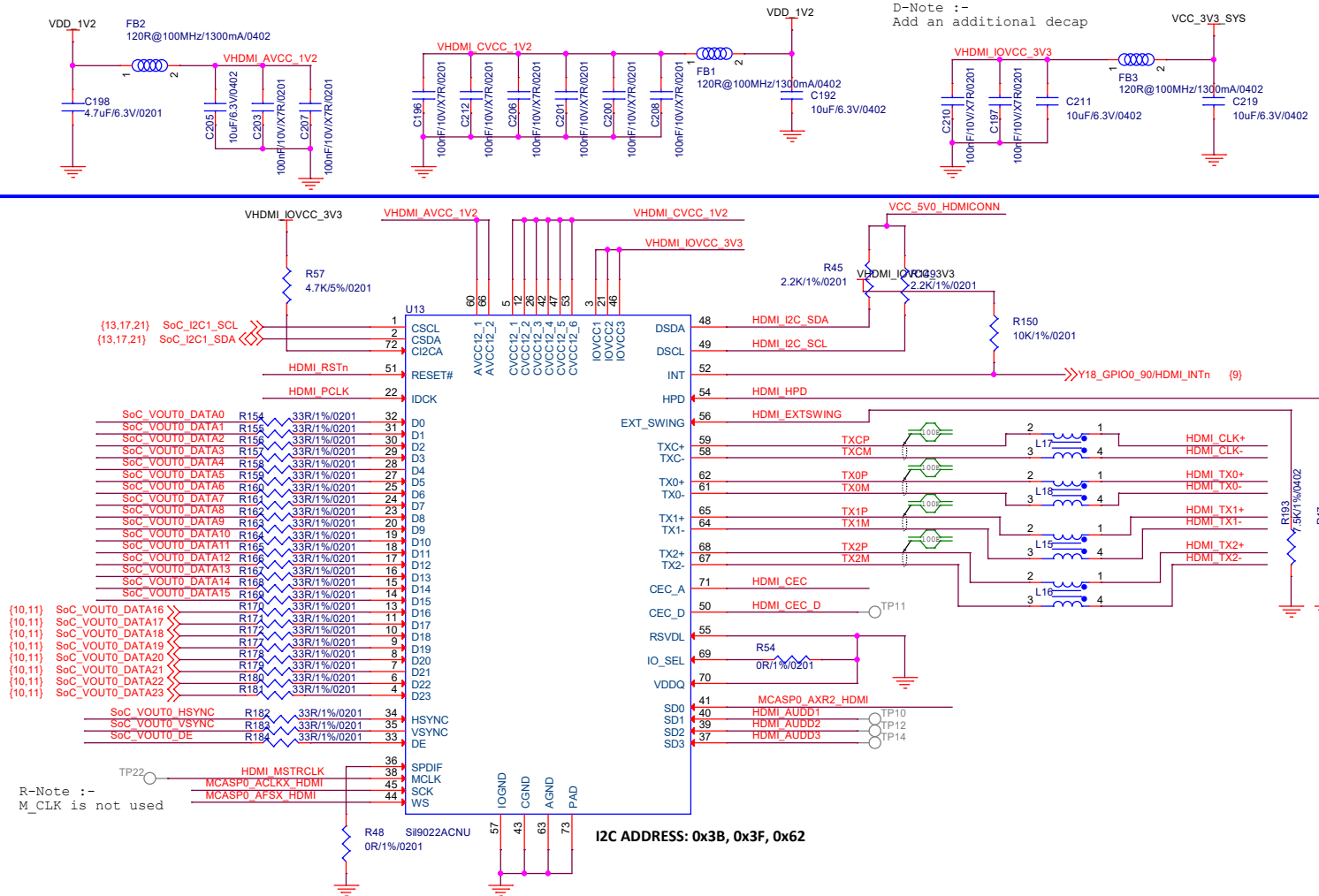
HDMI INTERFACE

U18M

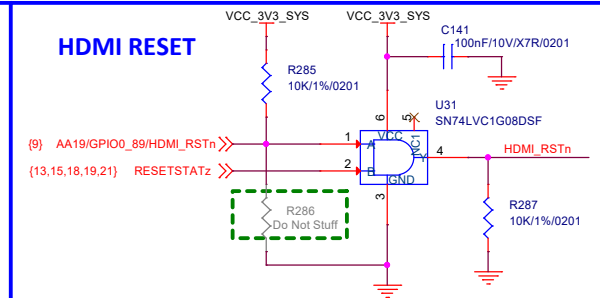
VOUT0_DATA0	U22	SoC_VOUT0_DATA0
VOUT0_DATA1	U21	SoC_VOUT0_DATA1
VOUT0_DATA2	U20	SoC_VOUT0_DATA2
VOUT0_DATA3	U19	SoC_VOUT0_DATA3
VOUT0_DATA4	T19	SoC_VOUT0_DATA4
VOUT0_DATA5	U18	SoC_VOUT0_DATA5
VOUT0_DATA6	V21	SoC_VOUT0_DATA7
VOUT0_DATA7	V19	SoC_VOUT0_DATA8
VOUT0_DATA8	V18	SoC_VOUT0_DATA9
VOUT0_DATA9	W22	SoC_VOUT0_DATA10
VOUT0_DATA10	W21	SoC_VOUT0_DATA11
VOUT0_DATA11	W20	SoC_VOUT0_DATA12
VOUT0_DATA12	W19	SoC_VOUT0_DATA13
VOUT0_DATA13	Y21	SoC_VOUT0_DATA14
VOUT0_DATA14	Y22	SoC_VOUT0_DATA15
VOUT0_DATA15		
VOUT0_PCLK	AA22	SoC_VOUT0_PCLK
VOUT0_DE	U17	SoC_VOUT0_DE
VOUT0_VSYNC	V17	SoC_VOUT0_VSYNC
VOUT0_HSYNC	T18	SoC_VOUT0_HSYNC

AM62A74AUMHIANFR

(9) MCASP0_ACLKX	MCASP0_ACLKX_HDMI
(9) MCASP0_AFSX	MCASP0_AFSX_HDMI
(9) MCASP0_AXR2	MCASP0_AXR2_HDMI



HDMI RESET



NOR

inhand Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	22. HDMI INTERFACE
Date:	Wednesday, April 08, 2026
Designed by:	Reviewed by:
Rev:	V1.0
Sheet:	22 of 22